CHEMISTRY ADAPTLY

PROPERTIES OF METALS & ALLOYS

KEY VOCABULARY

Alloy: a mixture of a metal with one or more other elements.

Delocalised electrons: free electrons that carry charge and heat.

Malleable: can be bent and shaped — layers slide.

Conductor: allows electricity/heat to pass through easily.

COMMON MISCONCEPTIONS

~~Alloys are compounds.~~

✓ Alloys are mixtures, not compounds.

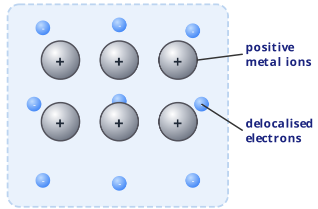
~~Alloys are softer than pure metals.~~

✓ Alloys are usually harder than pure metals.

~~Metal ions carry the current.~~

✓ Delocalised electrons carry the current.

METALLIC STRUCTURE



Different sized atoms in an **alloy** stop the layers sliding, making it harder.

1 CONDUCTIVITY

Explain why metals are good conductors of electricity. **[2 marks]**

2 DEFINE ALLOY

State what an **alloy** is. **[1 mark]**

3 WHY HARDER

Explain why an alloy is harder than the pure metal. **[3 marks]**

Think about the size of the atoms and the sliding layers.

4 MALLEABILITY

Explain why pure metals are malleable. **[2 marks]**

5 FILL THE GAPS

Complete the sentence using the word bank. **[2 marks]**

An alloy contains atoms of _____ sizes, which makes it _____ than the pure metal.

different | harder | same | softer

6 CHOOSE AN ALLOY

Name an alloy and state one use linked to its properties. **[2 marks]**

TIP: *Alloys are harder because different-sized atoms stop the layers sliding.*

✓ **Think: Why is pure gold too soft for everyday jewellery?**